

The TCP/IP Protocol Suite

A Visual Architecture & Blue Team Analysis Guide

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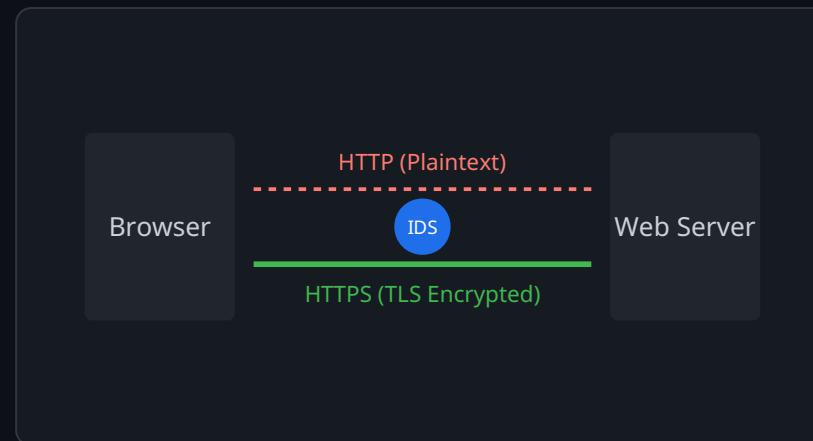
1. The Application Layer

The Application layer is where network data physically intersects with the software services users interact with. In modern cybersecurity, this layer is the primary battleground for data exfiltration and web-based exploitation.

DNS (Port 53) Translates human-readable domain names to IP addresses. Typically connectionless (UDP).

HTTP/S (Port 80/443) Facilitates web traffic. HTTPS encrypts the payload using TLS, rendering packet sniffing ineffective for payload inspection.

SNMP (Port 161) Used for querying hardware states (routers, switches) and configuring network devices remotely.



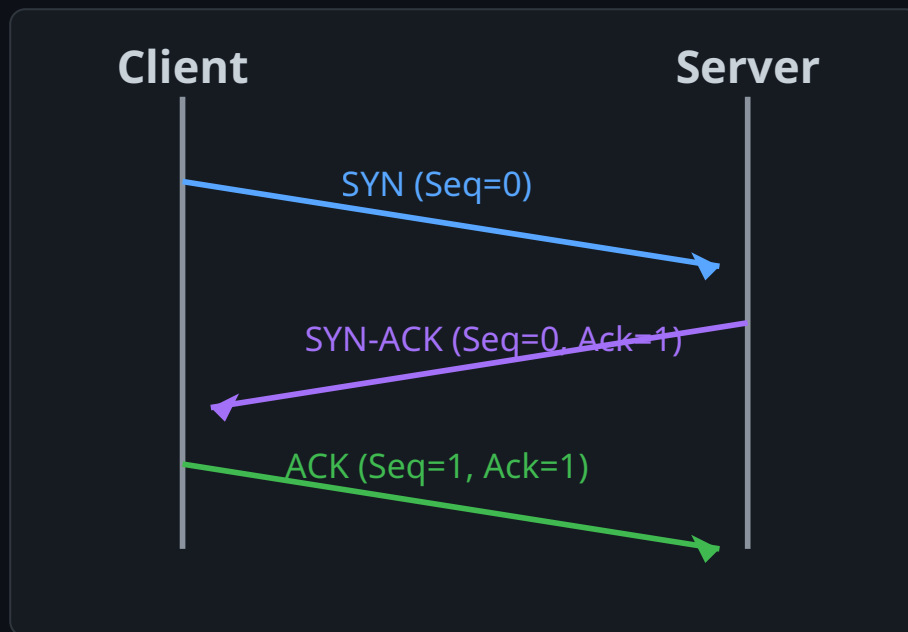
Blue Team Spotlight: DNS Tunneling

Because Port 53 is rarely blocked by enterprise firewalls, attackers often embed stolen data inside DNS queries to bypass egress filtering. Analysts must baseline normal DNS query length and volume. **Detection Logic:** Monitor for unusually long TXT records or high frequencies of unique subdomains originating from a single host.

2. The Transport Layer

This layer governs how connections are established and maintained. Understanding the mechanical differences between TCP (Reliable) and UDP (Fast) is critical for deciphering network scans and traffic floods.

TCP: The 3-Way Handshake



Blue Team Spotlight: Port Scanning

Adversaries utilize "Stealth SYN Scans" (like `nmap -sS`) to map networks. By sending a SYN packet and immediately dropping the connection when the server responds with SYN-ACK, they avoid completing the handshake, bypassing older application logs.

Wireshark Filter:

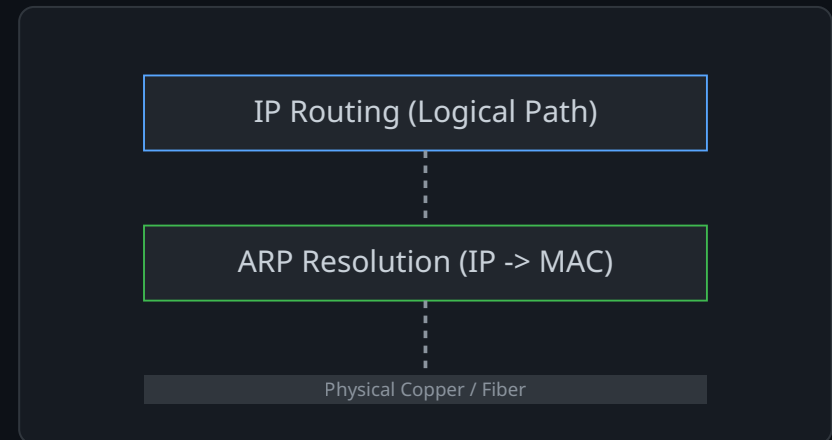
```
tcp.flags.syn == 1 and tcp.flags.ack == 0
```

3. The Internet & Network Access Layers

The lower layers handle the mechanics of routing data globally (IP) and delivering it locally via physical hardware (ARP & MAC Addresses).

Internet Layer (Layer 3): Relies on the Internet Protocol (IP) for logical addressing across subnets. The Internet Control Message Protocol (ICMP) operates here, providing vital diagnostic signaling.

Network Access (Layer 2/1): The Address Resolution Protocol (ARP) dynamically links an IP address to a physical Network Interface Card's MAC address inside a Local Area Network (LAN).



Blue Team Spotlight: ARP Poisoning

ARP lacks innate authentication. Attackers can flood a local switch with fraudulent "Gratuitous ARP" replies, tricking devices into associating the attacker's MAC address with the Default Gateway's IP address. This establishes a Man-in-the-Middle (MitM) position, allowing the interception of all outbound LAN traffic.

Analytical Summary Matrix

Protocol	Layer	Primary Function	SOC Monitoring Priority
HTTP / HTTPS	Application	Web Data Transfer	High (Payload inspection, malicious user-agents, SQLi attempts)
DNS	Application	Name Resolution	High (Data exfiltration, DGA domains)
TCP	Transport	Reliable Streaming	Medium (Identifying anomalous port scans or SYN floods)
ICMP	Internet	Diagnostics	Low/Medium (Detecting internal ping sweeps)
ARP	Network Access	IP to MAC mapping	High (Detecting local MitM positioning)